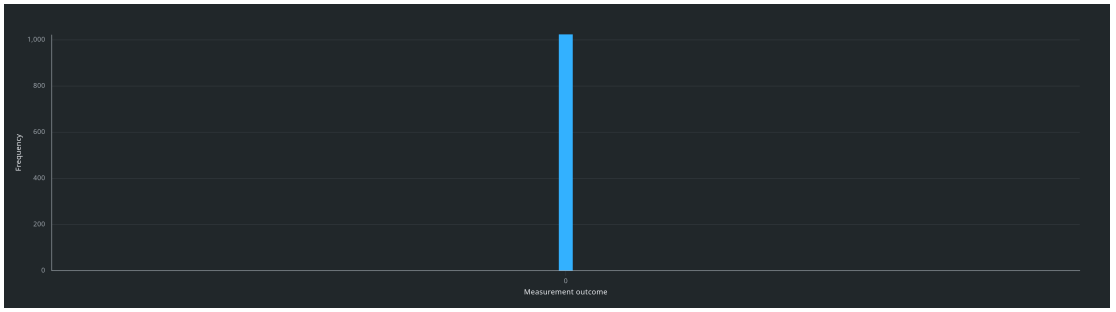
Lab 3 Meeting 2

PHYS:4340L Matthew Jarek

Task 1: Reproduce Basis Circuits

Circuit	|0⟩ %	|1⟩ %	|00⟩ %	|11⟩ %	Job ID
1A: All-Zero	100%	0%	NA	NA	d6aa80p7ce2c73fe20rg
1B: X-Flip	1.66%	98.34%	NA	NA	d6aatlp7ce2c73fe2pl0
1C: Bell	NA	NA	49.0%	48.1%	d6aa7l154hss73b60eg0

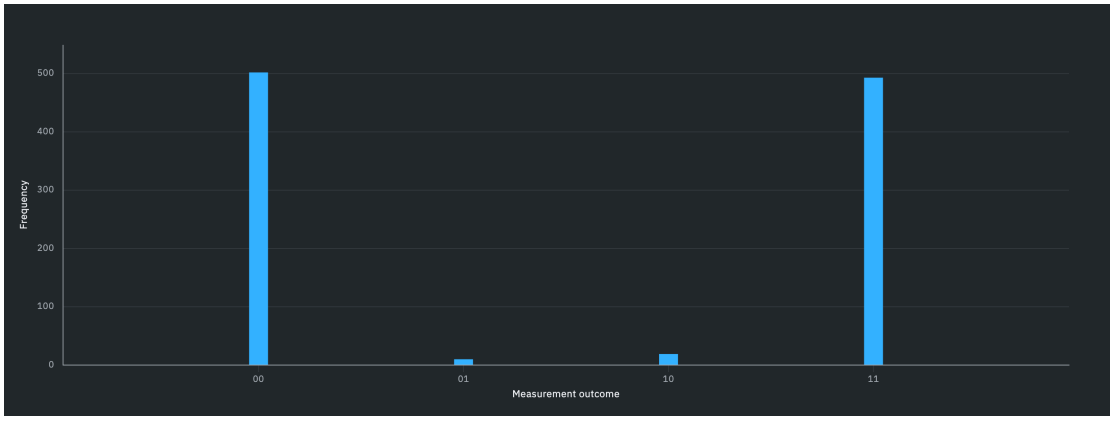
Circuit 1A: All-Zero



Circuit 1B: X-Flip



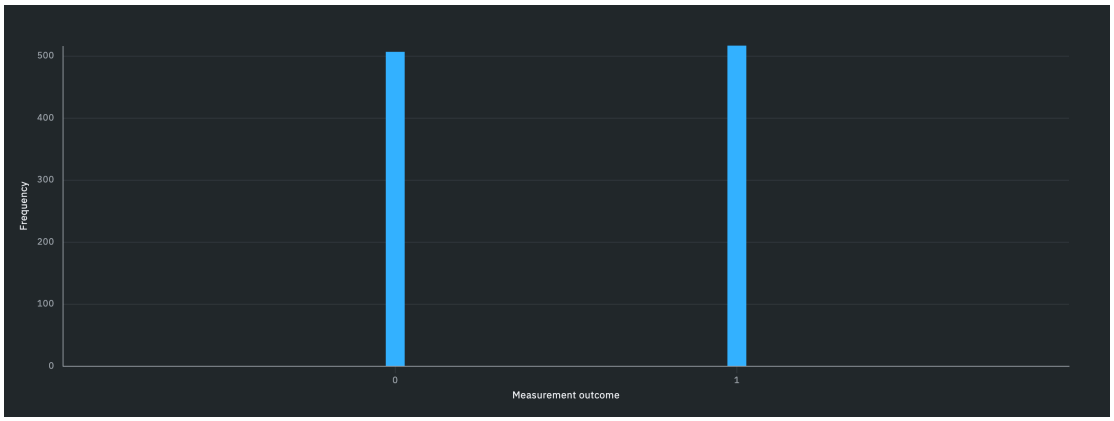
Circuit 1C: Bell



Task 2: Single-Qubit Modification

Part	Circuit	Predict	Actual	Why?
(a)	$\|0\rangle \rightarrow H \rightarrow X \rightarrow M$	50% $\|0\rangle$, 50% $\|1\rangle$	49.5% $\|0\rangle$, 50.5% $\|1\rangle$	Starting in $\|0\rangle$, applying the Hadamard then gives $1/\sqrt{2}\ \|0\rangle + 1/\sqrt{2}\ \|1\rangle$. Then applying the X gate gives us the same thing -> equal probability amplitudes -> equal probablities -> 50/50.
(b)	$\|0\rangle \rightarrow X \rightarrow H \rightarrow M$	50% $\|0\rangle$, 50% $\|1\rangle$	48.83% $\|0\rangle$, 51.17% $\|1\rangle$	Starting in $\|0\rangle$, applying the X gate then gives $\|1\rangle$. Then applying the Hadamard gate gives us $1/\sqrt{2}\ \|0\rangle - 1/\sqrt{2}\ \|1\rangle$ -> probability amplitudes -> equal probabilities -> 50/50.
(c)	$\|0\rangle \rightarrow X \rightarrow M \rightarrow H \rightarrow M$	50% $\|0\rangle$, 50% $\|1\rangle$	52.64% $\|0\rangle$, 47.36% $\|1\rangle$	Starting in $\|0\rangle$, applying the X gate then gives $\|1\rangle$, the measurement should return all $\|1\rangle$ (ignoring quatum error). Then applying the Hadamard gate gives us $1/\sqrt{2}\ \|0\rangle - 1/\sqrt{2}\ \|1\rangle$ -> probability amplitudes -> equal probabilities -> 50/50.

Circuit 2A: $|0\rangle \rightarrow H \rightarrow X \rightarrow M$



Circuit 2B: $|0\rangle \rightarrow X \rightarrow H \rightarrow M$



Circuit 2C: $|0\rangle \rightarrow X \rightarrow M \rightarrow H \rightarrow M$

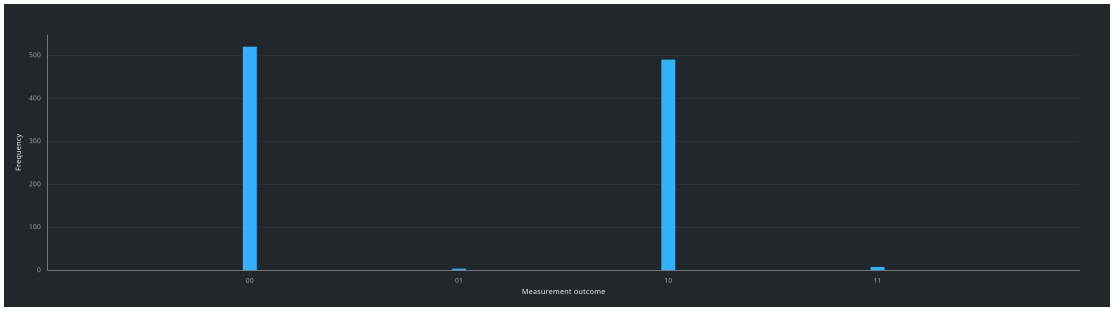


Task 3: Bell Circuit Variations

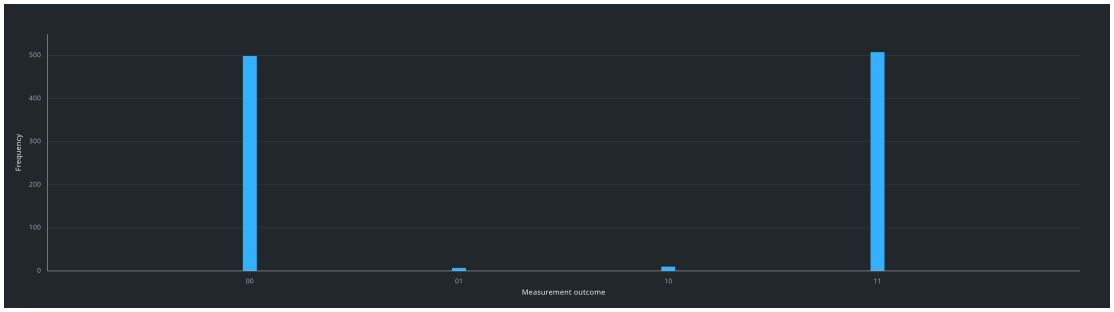
Circuit	$\|00\rangle$ %	$\|01\rangle$ %	$\|10\rangle$ %	$\|11\rangle$ %	Job ID

Swap H/CNOT	50.88%	00.39%	47.12	00.78	d6avgn4nsg9c7398cu2g
Measure q_0 first	48.73%	00.68%	00.98%	49.61%	d6avpp97ce2c73feq4sg
Z on q_1 pre-CNOT	51.46%	00.88%	00.49%	47.17%	d6avt4knsg9c7398dfag

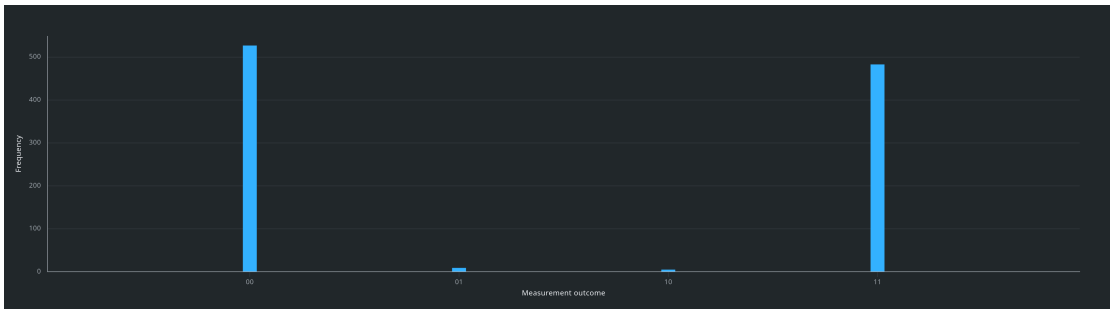
Circuit 3A: Swap H and CNOT



Circuit 3B: Measure q_0 first



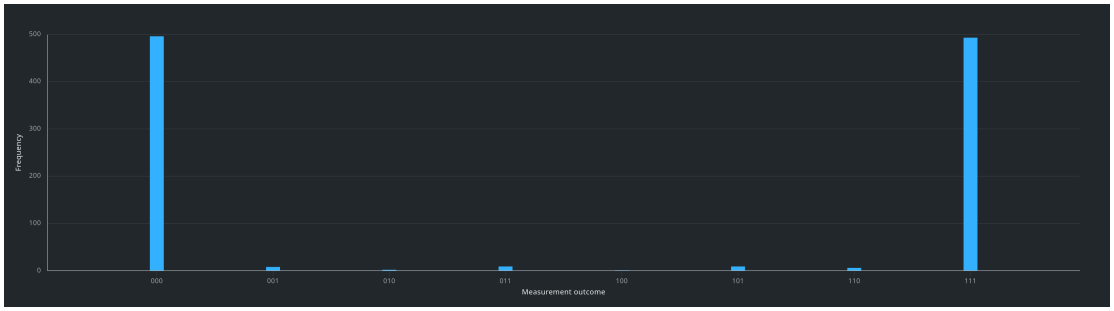
Circuit 3C: Z on q_1 before CNOT



Task 4: 3-qubit GHZ State

Circuit	**|000⟩ %**	**|001⟩ %**	**|010⟩ %**	**|011⟩ %**	**|100⟩ %**	**|101⟩ %**	**|110⟩ %**	**|111⟩ %**	
GHZ State	**48.44%**	**00.78%**	**00.20%**	**00.88%**	**00.10%**	**00.88%**	**00.59%**	**48.14%**	d6avu0nc

Circuit 4: 3-qubit GHZ State


500
400
300
200
100
0
Frequency
000
001
010
011
100
101
110
111
Measurement outcome